NORTH MAHARASHTRA'S

FIRST ROBOTIC CENTRE EQUIPPED WITH

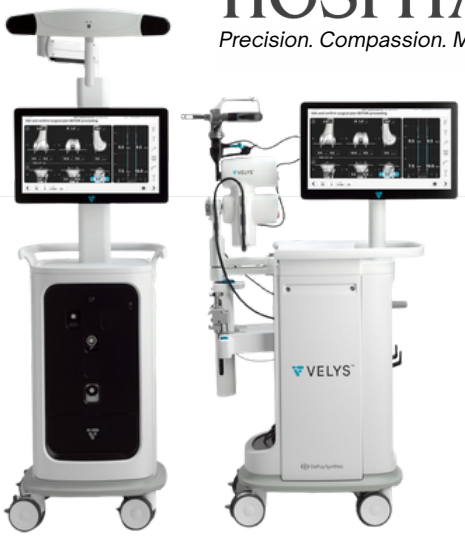
TOTAL KNEE REPLACEMENT

HIP REPLACEMENT

PARTIAL KNEE REPLACEMENT

TRUSTED HANDS ADVANCED TECHNOLOGY

DR. VISHAL KASLIWAL'S
KASLIWAL SUPERSPECIALITY HOSPITAL
Precision. Compassion. Motion.



HIP | TOTAL KNEE | PARTIAL KNEE REPLACEMENT
ALL IN ONE ROBOT

NAME :- _____

AGE/SEX :- _____

DIAGNOSIS :- _____

1st Floor Bosco Centre, Prasad Circle , Gangapur Road Nashik
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ABOUT US

Dr. Vishal Kasliwal has 15+ years of active practice with immense experience of robotic joint replacement with overall more than 5000 surgeries

North Maharashtra's first center with total hip, partial knee, total knee replacement with Johnson & Johnson VELYS™ Robotics for precise Knee & Hip Replacement surgeries.

Expert in knee-preserving procedures including High Tibial Osteotomy, cartilage restoration, meniscus repair, and ligament reconstruction surgeries.

Trained under Dr. Jin Park, with advanced expertise in shoulder and sports injury surgeries. Has worked alongside Dr. Mullaji and Dr. Tapasvi

Robotic Joint Replacement

Dr. Vishal Kasliwal

His commitment to precision-driven, minimally invasive joint replacement techniques including Subvastus Approach, Kinematic Alignment, Patient-Specific Alignment, and Direct Anterior Approach — focused on faster recovery, early rehabilitation, and improved patient comfort.

PRECISION REDEFINED WITH VELYS™ ROBOTIC SURGERY

Kasliwal Hospital brings advanced VELYS™ Robotic-Assisted Technology for precise and personalized joint replacement surgeries. This state-of-the-art system helps surgeons achieve better implant positioning, improved accuracy, and natural joint movement during Total Knee Replacement procedures.

VELYS™ Robotic Surgery offers benefits like smaller incisions, less pain, faster recovery, improved mobility, and shorter hospital stays. By combining cutting-edge robotic innovation with expert orthopedic care, Kasliwal Hospital ensures safer surgeries and better long-term outcomes, helping patients return to an active and pain-free lifestyle with confidence.

JOINTS FEEL NATURAL AGAIN WITH PERSONALISED AND INTUITIVE ROBOTIC KNEE REPLACEMENT

Knee osteoarthritis (OA) is a degenerative joint condition where the cartilage in the knee wears down, causing pain, stiffness, and reduced mobility.

Grade 0

Normal knee, no OA.

Grade 1

Minor cartilage wear, possible bone spur, no major symptoms.

Grade 2

Mild OA; small bone spurs, minor joint space narrowing, some pain.

Grade 3

Moderate OA; significant cartilage loss, joint space narrowing, pain, & stiffness.

Grade 4

Severe OA; almost no cartilage, large bone spurs, severe pain, and limited movement.

SYMPTOMS OF KNEE OA

- Pain in Daily Activities
- Grinding or cracking sound
- Weakness or instability
- Reduced range of motion

WHAT IS KNEE REPLACEMENT ?

Knee Replacement (TKR) is a surgical procedure where damaged knee joint surfaces are replaced with artificial implants to relieve pain and restore function.

A knee replacement consists of 3 basic steps:

1. Skin and soft tissue over the affected knee is incised. Muscles are retracted to gain access to underlying bone surfaces

2. Damaged cartilage coating at the end of femur (thigh bone) and tibia (shin bone) is removed manually or with robotic assistance

3. Removed cartilage and bone is replaced with metallic components - to recreate femoral and tibial surfaces. Metallic parts are cemented in place

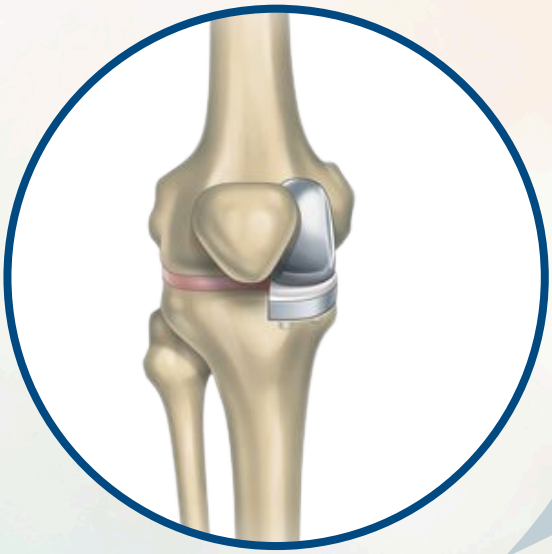
TYPES

TOTAL KNEE REPLACEMENT

This procedure has been revolutionary to orthopaedics giving many patients the ability to return to function, with implants lasting about 20-25 years. Physical therapy is tougher early on, and the knee sometimes doesn't quite feel the same as your own knee. Despite these potential limitations, patients have great patient satisfaction, high functional scores and longer-lasting replacements.

PARTIAL KNEE REPLACEMENT

Partial knee replacement is ideal for patients with pain on one side of the knee, strong ligaments, minimal deformity, and good knee movement. Compared to total knee replacement, recovery is usually quicker and easier, with less blood loss and lower risk of blood clots. If arthritis progresses to other parts of the knee it can later be successfully converted into a total knee replacement.



BENEFITS OF ROBOTIC TKR:

- Relieves chronic knee pain
- Improves walking ability
- Increases knee strength
- Enhances range of motion
- Corrects bow legs or knock knees
- Improves sleep by reducing night pain
- Boosts confidence in movement
- Allows return to daily activities
- Reduces dependency on painkillers
- Long-term solution (lasts 15–20+ years)
- Higher Precision
- Less Tissue Damage
- Smaller incision.
- Preserves natural ligaments
- Personalised approach

WHAT TO EXPECT DURING YOUR STAY HERE FOR JOINT REPLACEMENT

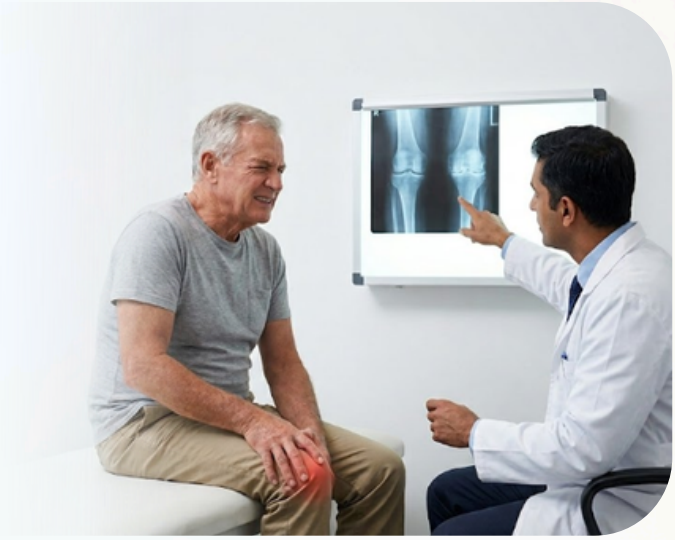
- Informed care designed for confident recovery
- Clear communication with personalized patient guidance
- Compassionate support at every stage of treatment
- Advanced diagnostics & robotic-assisted joint replacement
- Precision-driven surgery focused on comfort and mobility
- Dedicated rehabilitation & faster recovery support
- Expert multidisciplinary team led by Dr. Vishal Kasliwal
- Seamless care before, during & after surgery

All services are delivered at Kasliwal Super Speciality Hospital.

Advanced Technology.
Personalized Care.
Confident Recovery.

WHEN DOES A PATIENT KNOW THAT HE IS A CANDIDATE FOR KNEE REPLACEMENT?

Severe and persistent pain even at rest and moderate activities Severe stiffness Failure of conservative treatment Knee deformity.



PRE-OPERATIVE INVESTIGATION THE DOCTOR MIGHT ADVISE PRIOR TO SURGERY

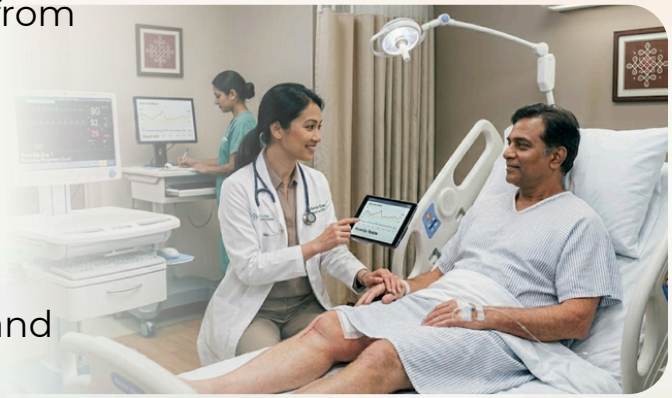
- Blood test
- Urine routine and culture
- Chest X-ray
- Hip-to-ankle scannogram X-ray
- ECG 2D echo or stress ECHO
- Physician fitness and cardiac fitness

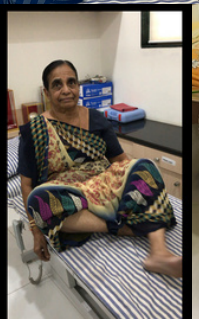
YOUR SURGERY AND HOSPITAL STAY:

- Admission 1 day prior to surgery.
- You will be kept nil by mouth since midnight and necessary preparations will be done.
- On day 2 you will be posted for surgery.
- Single TKR usually takes one hour but the actual elapsed time may vary.
- Generally surgery is done under spinal anaesthesia.

- - Patient will be shifted to ICU with Foley's catheter after surgery and post-op X-ray.
- - ICU observation for 12–16 hours to monitor vital signs.
- - IV antibiotics, analgesics, antacids, antiemetics, and epidural anaesthesia will be administered.
- - If required, a femoral block may be given by the anaesthetist for better pain relief.

- Nill by mouth for 4–6 hours post-surgery.
- Gradual shift from liquids to normal diet by Day 2.
- Physiotherapy and mobilisation twice daily from Day 2.
- Catheter removal after mobilisation.
- Dressing and discharge on Day 3 with medications and crepe bandage.
- Round-the-clock monitoring of vitals, pain, and intake/output.
- Daily review by Dr. Vishal Kasliwal.





Instructions After Discharge

Take medications as advised

Do not wet dressing

Contact immediately for fever, chills, or dressing soakage.

Physiotherapy should be done at least twice daily

Have healthy protein and calcium rich diet

Excessive soakage or unusual symptoms should be reported immediately. Early medical attention helps prevent complications.

Possible Risks and Complications

- Infection
- Nerve or blood vessel damage
- Blood clot
- Stiffness or limited movement
- Implant failure or loosening
- Persistent pain
- Allergic reaction

What will be your exercise protocol ?

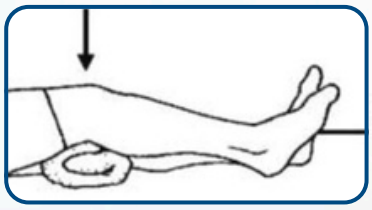
Immediate(Day 0 to Day 3)

- Goal- minimum 0-80° knee ROM should be achieved.
- Icing 3-4 times a day.
- Limb elevation to minimize swelling.
- Introduction to acute postoperative exercise program.
- Weight bearing as tolerated (WBAT) with assistive device to full weight-bearing. .
- Use of CPM (continuous passive machine) for knee ROM, if necessary.

Exercises

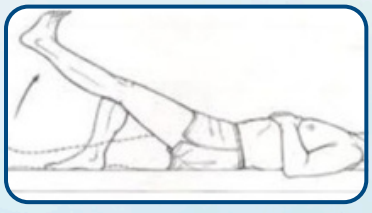
1.Ankle pumps

2. Quadriceps strengthening exercises



I. Static quadriceps

ii. VMO strengthening (short arc strengthening)



iii. Straight leg raises

3. Heel slides (Passive ROM to Active assisted ROM to Active ROM as tolerated)

4. Gluteal sets

Scan the above qr code for Excercise Video

Phase 2 (Day3- Week 6)

- Improve knee ROM from 90–120° and achieve full extension.
- Functional transfer training: sit-to-stand, toilet transfers, bed mobility.
- Start stationary cycling (5 mins) after week 2.
- Use NMES for quads if contraction is poor (parameters based on exercise goal).
- Gait training to improve swing-through and stance phase; aim to discontinue assistive device by end of week 2.
- Proprioception and balance training.

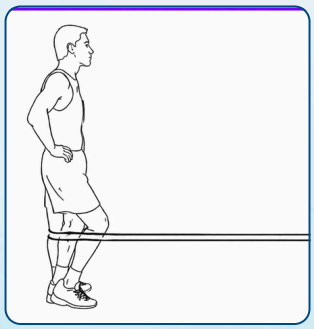
Exercises

1. Quadriceps sets, Quadriceps long arc and short arc strengthening exercises

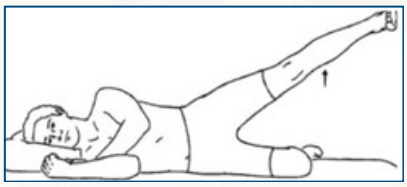
2. Gluteal sets

3. Straight leg raises

4. Standing terminal knee extension exercises with theraband

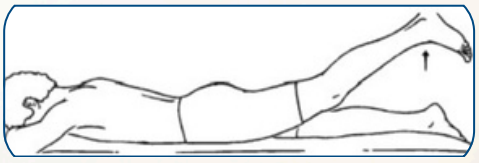


5. Hip ROM exercise- Straight leg raises in all 3 planes

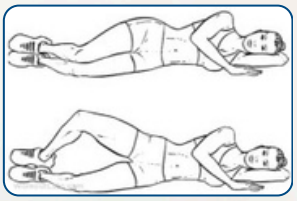


i. Straight leg raises in supine

ii. Hip abduction



iii. Hip extension in prone



iv. Clamshells

Phase 3 (week 7-12)

Exercises

1. Bridging with progression (unilateral or with unstable surface or on physioball)

2. Proprioceptive neuromuscular facilitation exercises with theraband or manual resistance.

3. Progression of balance exercises

Phase 4 (week 12 - week 16)

- Return to appropriate recreational sports / activities as indicated
- Enhance strength, endurance and proprioception as needed for activities of daily living and recreational activities
- Continue previous exercises with progression of resistance and repetitions.
- Increased duration of endurance activities.
- Initiate return to specific recreational activity: golf, doubles tennis, progressive walking or biking program

Note -

Total knee replacement can enhance your quality of life by providing many years of improved mobility and reduced pain. According to the survey conducted by TOI in 2022 each year, over 2 lakh Indians benefit from total knee replacement. In the last decade, remarkable advances in technology have transformed total knee replacement into an efficient and widely performed procedure.

KASLIWAL SUPERSPECIALITY HOSPITAL

Better Care, Better Life

IN CASE OF EMERGENCY OR QUERIES

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